

PGR106

Linear Algebra

Lecture 3 – Matrix Determinant and Inverse



Determinant

- **Determinant** is a function, mapping matrix to real scalars (**$\det(\mathbf{M})$**).
- **Determinant** can be computed only for square matrices.
- **Determinant**
 - Solving Systems of Linear Equations
 - Eigenvalues and Eigenvectors
 - Principal Component Analysis (PCA)

Determinant

Example

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
$$2 \times 2$$

$$\det(M) = |M| = a * d - b * c$$

$$M = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$
$$2 \times 2$$

$$\begin{aligned} \det(M) &= 5 * 2 - 1 * 4 \\ &= 10 - 4 \\ &= 6 \end{aligned}$$

Determinant

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

3×3

Checker Board

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$|M| = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$|M| = a(e * i - f * h) - b(d * i - f * g) + c(d * h - e * g)$$

Determinant

Example

$$M = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 3 & -1 & 4 \end{bmatrix}$$

$$|M| = 2 \begin{vmatrix} 5 & -3 \\ -1 & 4 \end{vmatrix} - 3 \begin{vmatrix} -1 & -3 \\ 3 & 4 \end{vmatrix} + 4 \begin{vmatrix} -1 & 5 \\ 3 & -1 \end{vmatrix}$$

$$|M| = 2(5 * 4 - (-3 * -1)) - 3(-1 * 4 - (-3 * 3)) + 4(-1 * -1 - 5 * 3)$$

$$|M| = 2(20 - 3) - 3(-4 - (-9)) + 4(1 - 15)$$

$$2(20 - 3) - 3(-4 + 9) + 4(1 - 15)$$

$$2(17) - 3(5) + 4(-14) = 34 - 15 - 56 = -37$$

Determinant

Determinant of 2 x 2 matrix:

$$|M| = a * d - b * c$$

Determinant of 3 x 3 matrix:

using $|M|$ of 2×2 submatrices

Determinant of 4 x 4 matrix:

using $|M|$ of 3×3 submatrices

Determinant of n x n matrix:

using $|M|$ of $(n - 1) \times (n - 1)$ submatrices

Determinant

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

Its better to select a row or column of matrix which contains many zeros

Determinant: can be positive, negative or zero

A matrix with Determinant as zero is a **singular matrix**

Properties of Determinant

$$\det(I) = 1$$

$$\det(A) = \det(A^T)$$

$$\det(A) = \frac{1}{\det(A^{-1})}$$

$$\det(AB) = \det(A) * \det(B)$$

$$\det(A^n) = \det(A)^n$$

Matrix Inverse

- **Matrices** are tabular representation of data arranged in rows and columns.
- **Addition** is performed between matrices of same dimensions.
- Matrix **multiplication** is performed based on rules of matrix multiplication.
- Instead of matrix division, matrix **inverse** is computed.
- **Inverse**
 - Solving Linear Systems
 - Linear Regression

Matrix Inverse

⑦

Inverse of 7 = $\frac{1}{7} = 7^{-1}$

$$7 * \frac{1}{7} = 1$$

$$7 * 7^{-1} = 1$$

$$M = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix}$$

$n \times n$

$$M^{-1}M = I$$

$$MM^{-1} = I$$

Matrix Inverse

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

2×2

$$M^{-1} = \frac{1}{\det(M)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det(M) \neq 0$$

Matrix Inverse

- Only **square** matrices can have inverse.
- Every square matrix **may not** have an inverse.
- Inverse of a square matrix is **unique**.

Matrix Inverse

Example

$$M = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\det(M) = 2 * 2 - 3 * 1 = 4 - 3 = 1$$

$$M^{-1} = \frac{1}{1} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

Matrix Inverse

Example

$$M = \begin{bmatrix} 4 & 6 \\ 1 & 2 \end{bmatrix}$$

$$\det(M) = 4 * 2 - 6 * 1 = 8 - 6 = 2$$

$$M^{-1} = \frac{1}{2} \begin{bmatrix} 2 & -6 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ -0.5 & 2 \end{bmatrix}$$

Singular and Invertible Matrices

- Inverse of a matrix M exists if the $\text{det}(M)$ is not equal to zero.
- If inverse exists, then the matrix is said to be **invertible**.
- An **invertible** matrix is also called **non-singular matrix**.

Singular and Invertible Matrices

- If $\det(M) = 0$, then matrix M does not have an inverse.
- If no inverse exists, then the matrix is said to be non-invertible.
- A non-invertible matrix is also called singular matrix.

Properties of Matrix Inverse

$$A^{-1}A : \textit{Left Inverse}$$

$$A A^{-1} : \textit{Right Inverse}$$

$$A^{-1}A = A A^{-1}$$

$$(A^{-1})^{-1} = A$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^T)^{-1} = (A^{-1})^T$$

End of Lecture 3



MATHEMATICS FOR MACHINE LEARNING

Marc Peter Deisenroth
A. Aldo Faisal